16.1 : Answers

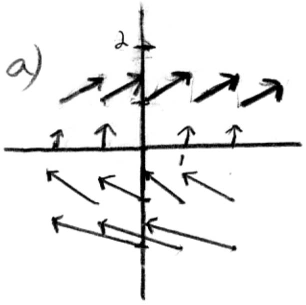
1. a) II   b) IV   c) I   d) III

see explanation below ☆

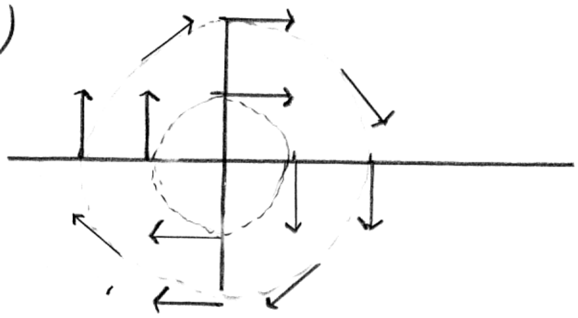
2. a) IV   b) I   c) III   d) II

☆☆ see explanation below

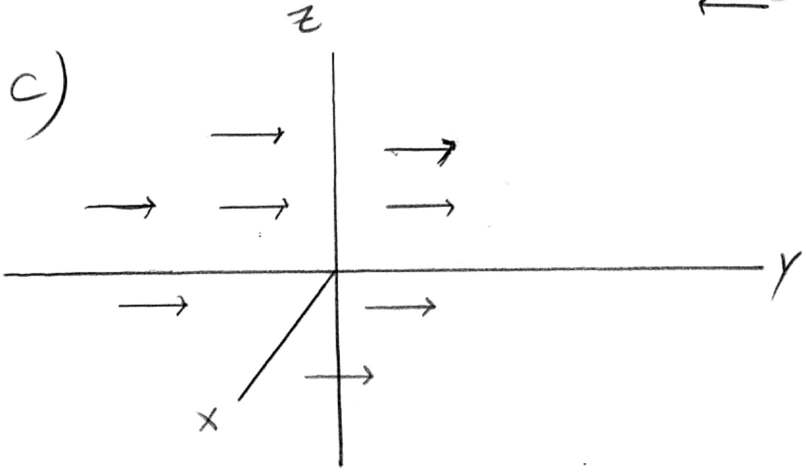
3. a)



b)



c)



4. a) $\nabla f(x,y) = \dfrac{1}{x+2y}\,\vec{i} + \dfrac{2}{x+2y}\,\vec{j}$

b) $\nabla f(x,y,z) = \cos\left(\dfrac{y}{z}\right)\vec{i} - \dfrac{x}{z}\sin\left(\dfrac{y}{z}\right)\vec{j} + \dfrac{xy}{z^2}\left(\sin\dfrac{y}{z}\right)\vec{k}$

☆ Since x is constant, vectors along horizontal lines (i.e. when y is constant) should be identical. Along vertical lines, since $\vec{j}$-component is $\sin y$, vectors should eventually repeat themselves. Only graph IV satisfies these conditions.

☆☆ For $\vec{F} = \vec{i} + 2\vec{j} + z\vec{k}$, look for the graph in which the $\vec{i}, \vec{j}$ components are constants. Since the $\vec{k}$ component is z, vectors above the xy-plane point up, while vectors below the xy-plane point down (vectors in the xy-plane are constant). Only graph I satisfies these conditions.